2.7. BASIC RULES

Apart from the fundamental laws, some rules are also helpful to simplify the circuits.

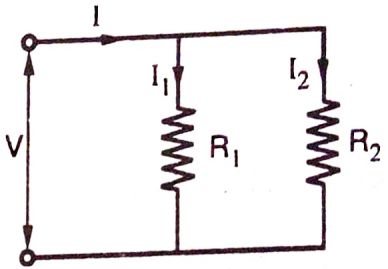


Fig. 2.28. Parallel Circuit.

2.7.1. Current division rule. Current is divided along the parallel branches while voltage remains same across them. Let us consider the parallel circuit of Fig. 2.28.

By KCL
$$I = I_1 + I_2 \qquad ...(2.35)$$

$$I_1 = \frac{V}{R_1}$$

and
$$I_2 = \frac{V}{R_2}$$

But
$$V = I \times R_{eq} = I \times \frac{R_1 R_2}{R_1 + R_2}$$

So,
$$I_1 = I \times \frac{R_2}{R_1 + R_2} \qquad ...(2.36)$$

and
$$I_2 = I \times \frac{R_1}{R_1 + R_2} \qquad ...(2.37)$$

2.7.2. Voltage Division Rule. Voltage is divided in series circuit while current remains same. Fig. 2.29 shows the series circuit with two resistance R_1 and R_2

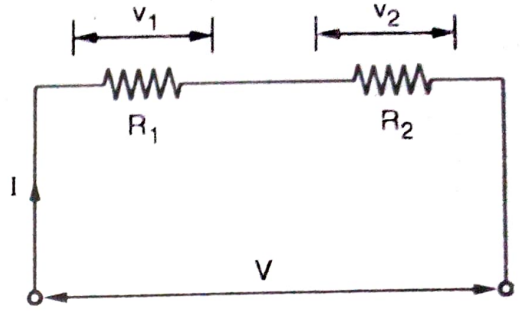


Fig. 2.29. Series Circuit.

By KVL $$V = V_1 + V_2 \qquad ...(2.38)$$

$$V_1 = R_1 I$$

and $$V_2 = R_2 I$$

But $$I = \frac{V}{R_1 + R_2}$$

So,
$$V_1 = V \times \frac{R_1}{R_1 + R_2} \qquad ...(2.39)$$

and
$$V_2 = V \times \frac{R_2}{R_1 + R_2} \qquad ...(2.40)$$

2.7.3. Source Transformation (Source Conversion). A voltage source in series with a resistor can be converted into a current source with the parallel resistor and vice-versa.

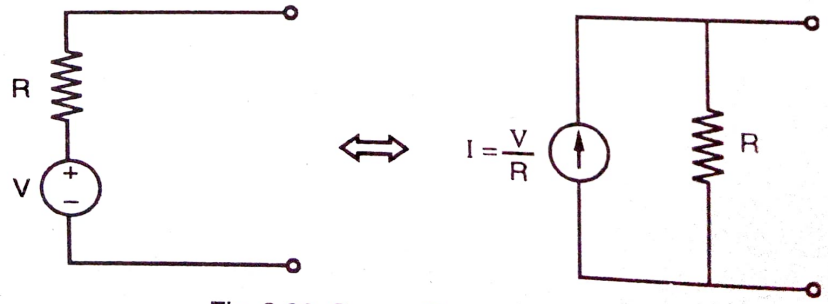


Fig. 2.30. Source Transformation

Fig. 2.30. Source Transformation.

2.8. SOME MORE RULES ABOUT SOURCES

(*i*) Series voltage sources are added.

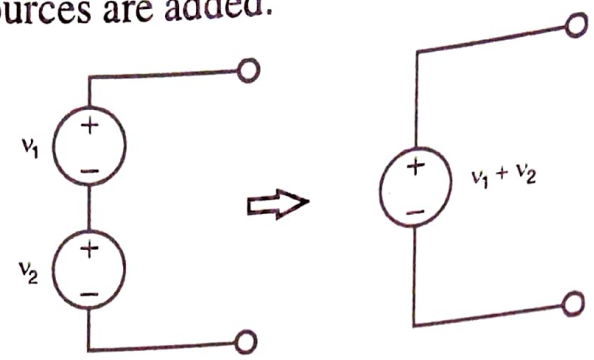


Fig. 2.31.

(*ii*) Parallel current sources are added.

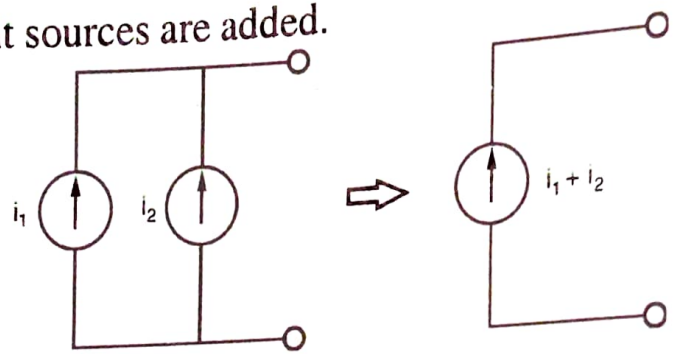


Fig. 2.32.

(*iii*) If two or more voltage sources are connected in parallel, then they must be identical otherwise large circulating current may damage the circuit.

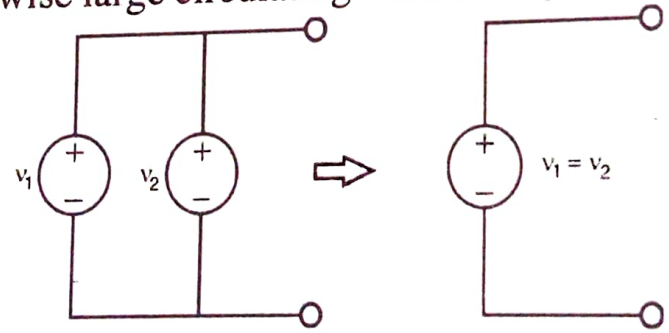


Fig. 2.33.

(*iv*) If two or more current sources are in series, they must have the identical value.

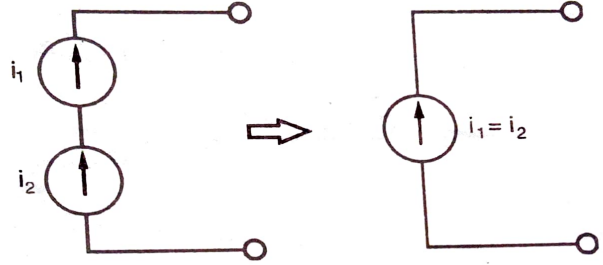


Fig. 2.34.

EXAMPLES ON CALCULATION OF EQUIVALENT R, L, C

Example 2.1. *Find the equivalent resistance for the circuit shown in Fig. 2.35.*

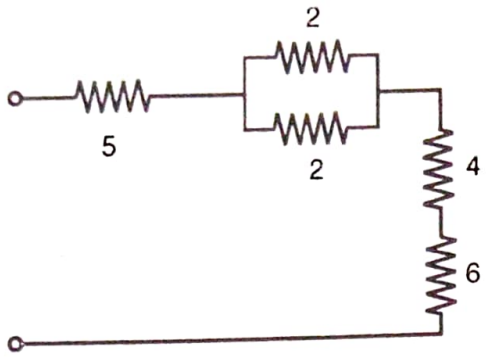


Fig. 2.35.

Solution. After adding the series resistances, the circuit becomes, as shown in Fig. 2.36.

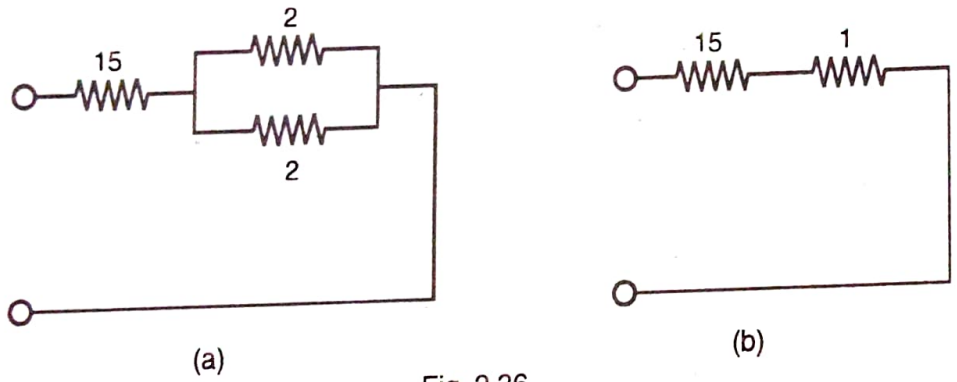


Fig. 2.36.

$$R_{se} = 5 + 4 + 6 = 15\ \Omega$$

Now solve the parallel branch,

$$R_{\text{parallel}} = 2 \parallel 2 = \frac{2 \times 2}{2 + 2} = 1\ \Omega$$

$$R_{eq} = R_{se} + R_{\text{parallel}} = 15 + 1 = 16\ \textbf{Ans.}$$

Example 2.2. *Find the equivalent capacitance for the Fig. 2.37.*

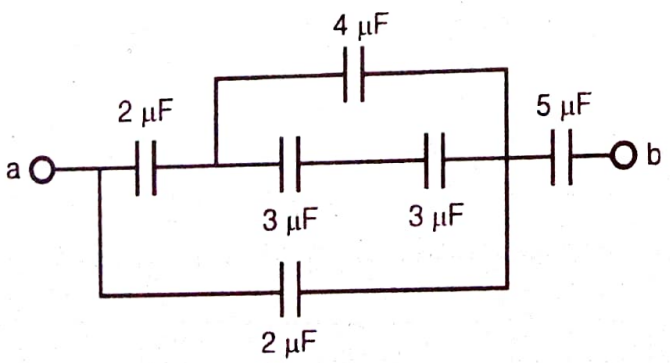


Fig. 2.37.

Solution. First, we shall solve the series combination of 3 and 3 μF capacitors.

$$= \frac{3 \times 3}{3+3} = \frac{9}{6} = 1.5\ \mu F$$

This is in parallel with 4 μF $= 1.5 + 4 = 5.5\ \mu F$

Now this is in series with 2 μF $= \frac{5.5 \times 2}{5.5+2} = \frac{11}{7.5} = 1.466\ \mu F$

Again this 1.466 μF is in parallel with 2 μF

$$1.466 + 2 = 3.466\ \mu F$$

3.466 μF is in series with 5 μF $C_{eq} = \frac{3.466 \times 5}{3.466+5} = 2.047\ \mu F.$

Example 2.3. *Determine the equivalent inductance of circuit shown in Fig. 2.38.*

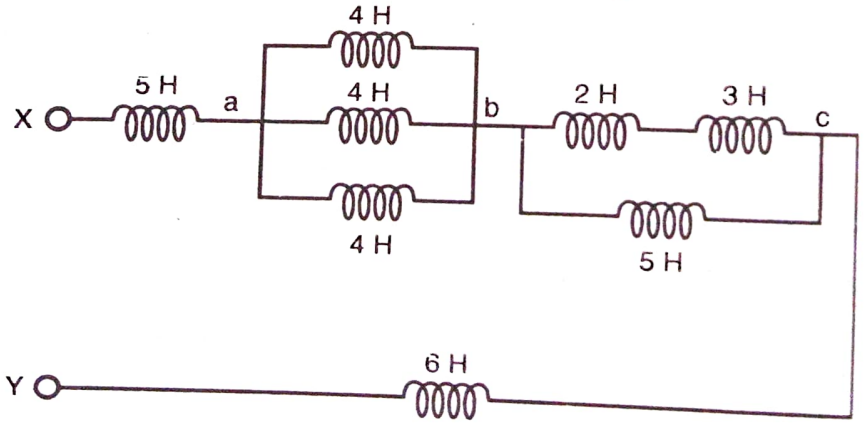


Fig. 2.38

Fig. 2.38.

Solution. First of all we shall solve the parallel branches between points a and b

$$\frac{1}{L_{ab}} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$$

$$L_{ab} = \frac{4}{3} = 1.333\ H$$

Now, between point b and c, $2\ H$ and $3H$ are in series and they are in parallel with 5 H.

$$L_{bc} = \frac{(2+3)\,5}{2+3+5} = \frac{5}{2} = 2.5\ H\,.$$

The given circuit becomes :

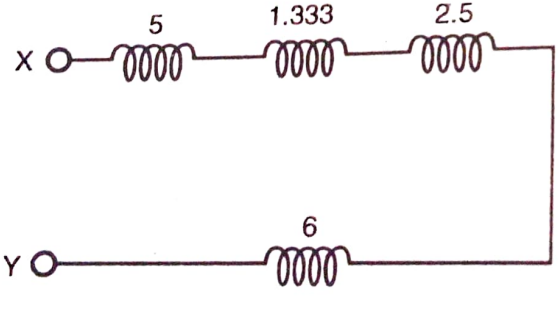


Fig. 2.39.

$\therefore$ $L_{xy} = 14.833\ H.$

Example 2.4. *Determine the equivalent capacitance seen from the source. All the capacitances are in Farad.*

Solution. $5 \parallel 5 \parallel 10 = 5 + 5 + 10 = 20\ \text{F}$

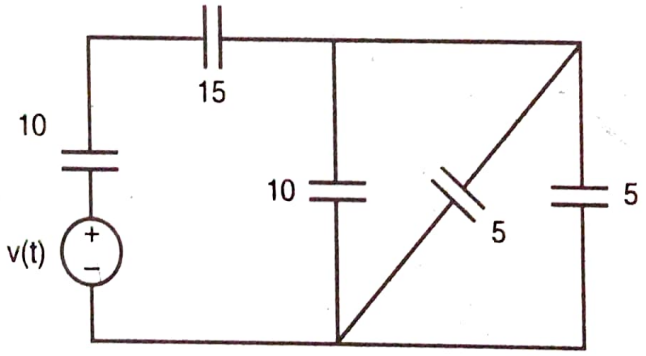


Fig. 2.40.

Now 10F, 15F and 20F are in series,

$$\frac{1}{C_{eq}} = \frac{1}{10} + \frac{1}{15} + \frac{1}{20}$$

$$= 0.1 + 0.066 + 0.05$$

$$C_{eq} = 4.629\ \text{F}.$$